

ORIGINAL ARTICLE

MRAs and Race in Heart Failure: An Individual Participant Data Meta-Analysis of Randomized Trials

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BACKGROUND: There are concerns that renin-angiotensin system inhibitors are less effective in Black patients than non-Black patients with heart failure (HF). We examined the efficacy and safety of mineralocorticoid-receptor antagonists (MRAs), compared with placebo, in patients with HF with reduced ejection fraction or HF with mildly reduced/preserved ejection fraction, according to self-reported race (Black or non-Black).

METHODS: This was a post hoc individual participant data meta-analysis of the 4 large placebo-controlled trials comparing MRAs to placebo in patients with HF with reduced ejection fraction (RALES [Randomized Aldactone Evaluation Study], EMPHASIS-HF [Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure]) and HF with mildly reduced/preserved ejection fraction (TOPCAT [Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist], FINEARTS-HF [Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure]). The primary outcome was a composite of cardiovascular death or first HF hospitalization.

RESULTS: Of the 13846 patients randomized in the 4 trials, 577 (4.2%) identified as Black. Despite being younger (64 versus 70 years), rates of HF hospitalizations and death were higher in Black than non-Black patients. The hazard ratio for MRA versus placebo for the primary composite outcome was 0.87 (95% CI, 0.66–1.15) in Black patients and 0.77 (95% CI, 0.72–0.82) in non-Black patients ($P_{\text{interaction}}=0.34$), with 3.9 and 2.7 fewer events per 100 person-years of treatment, respectively. For first HF hospitalization, the hazard ratios were 0.86 (95% CI, 0.63–1.17) and 0.73 (95% CI, 0.68–0.80) for Black and non-Black patients, respectively ($P_{\text{interaction}}=0.36$). The corresponding hazard ratios for cardiovascular death were 0.75 (95% CI, 0.48–1.17) and 0.81 (95% CI, 0.74–0.90), respectively ($P_{\text{interaction}}=0.80$). Adverse events with MRAs, compared with placebo, were not modified by race. The effects of MRAs in patients with HF with reduced ejection fraction and HF with preserved ejection fraction, individually, were not modified by race.

CONCLUSIONS: There was no statistically significant evidence of heterogeneity in the absolute or relative effects of MRAs on clinical outcomes between Black and non-Black patients with HF, regardless of HF phenotype.

Key Words: angiotensin-converting enzyme ■ heart failure ■ hospitalization ■ mineralocorticoid-receptor antagonists ■ phenotype

Whether the efficacy of mineralocorticoid-receptor antagonists (MRAs) in heart failure (HF) varies by race is uncertain. However, compared with White individuals, people of African ancestry more

often have low plasma renin activity but similar plasma aldosterone concentrations; that is, aldosterone levels may be inappropriately high for their renin and sodium/volume status.^{1–4} Also, a stronger association between

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This manuscript was sent to Onyedika J. Ilonze, MD, MPH, Guest Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCHEARTFAILURE.125.014355>.

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Circulation: Heart Failure is available at www.ahajournals.org/journal/circheartfailure

WHAT IS NEW?

- In a patient-level pooled meta-analysis of almost 14 000 participants from large placebo-controlled clinical trials, mineralocorticoid-receptor antagonists reduced the absolute and relative risk of heart failure (HF) hospitalization (both first and total HF hospitalizations), cardiovascular death, and all-cause death with consistent effects in Black and non-Black patients.
- The effects of mineralocorticoid-receptor antagonists on experiencing hypotension, hyperkalemia, hypokalemia, or elevated creatinine levels did not differ meaningfully by race. The benefits of mineralocorticoid-receptor antagonists in HF with reduced ejection fraction and HF with preserved ejection fraction were consistent irrespective of race.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Although this meta-analysis combined all large placebo-controlled mineralocorticoid-receptor antagonist trials in HF, the representation of Black patients was modest, reflecting their underrepresentation in clinical trials. Adequate inclusion of Black patients (and other underrepresented groups) is necessary to ensure the generalizability of study findings to all patients with HF.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
eGFR	estimated glomerular filtration rate
EMPHASIS-HF	Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure
FINEARTS-HF	Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
HR	hazard ratio
LVEF	left ventricular ejection fraction
MRA	mineralocorticoid-receptor antagonist
NT-proBNP	N-terminal pro-B-type natriuretic peptide
RALES	Randomized Aldactone Evaluation Study
SBP	systolic blood pressure
TOPCAT	Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist

aldosterone concentrations and blood pressure has been reported in Black compared with White patients.^{1–4} In addition, in a clinical study, the mineralocorticoid-agonist 9-alpha fludrocortisone increased blood pressure in Black but not White participants.⁵ Whether these findings have genetic, environmental, or socioeconomic determinants is unknown, but they do raise the possibility that the therapeutic response to MRAs might vary by race in HF.⁶ In a post hoc analysis of RALES (Randomized Aldactone Evaluation Study), Black patients with HF with reduced ejection fraction (HF_rEF; n=120) appeared to derive less clinical benefit from spironolactone than non-Black patients.⁷ In contrast, no heterogeneity in the effect of spironolactone was observed in Black patients with HF with mildly reduced ejection fraction/HF with preserved ejection fraction (HF_pEF; n=302), compared with other participants, enrolled in TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) in the Americas, although this trial was neutral overall,⁸ and the number of patients and events in RALES and TOPCAT individually were modest.^{9,10} Consequently, to get a more robust estimate of the efficacy and safety of MRAs according to race, we have conducted an individual participant data meta-analysis of the 4 large placebo-controlled trials comparing MRAs to placebo in patients with HF. These trials included EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure) in patients with HF_rEF and FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure) in participants with HF with mildly reduced ejection fraction/HF_pEF, in addition to RALES and TOPCAT.^{9–12} Specifically, we provide a detailed report of the effects of MRAs on clinical outcomes, including mortality and HF hospitalization (both first and total hospitalizations), along with a description of safety, in Black compared with non-Black patients across the range of left ventricular ejection fraction (LVEF).

METHODS

Search Strategy and Selection Criteria

The corresponding author had full access to all the trial data and takes responsibility for its integrity and the data analysis. Individual participant data from 4 large double-blind randomized placebo-controlled trials, RALES, EMPHASIS-HF, TOPCAT, and FINEARTS-HF, were included in the present study. The trial protocols were approved by the ethics committee at all participating institutions, and all patients provided written informed consent.^{9–12} To ensure that we did not exclude any other important trials, we conducted a systematic review of MEDLINE via PubMed of randomized trials of MRAs in patients with HF, published from database inception to June 1, 2024. The search strategy is detailed in [Supplemental Methods](#). Trials were included if they enrolled at least 1000 patients with HF with an appropriately powered morbidity or mortality outcome. No further trials were identified from the search ([Figure S1](#)).

In RALES, 1663 ambulatory patients with a LVEF $\leq$ 35% and New York Heart Association class III to IV, treated with an ACE (angiotensin-converting enzyme) inhibitor (if tolerated) and a loop diuretic, were randomly assigned to spironolactone or matching placebo. Patients with a serum creatinine level >2.5 mg/dL or a serum potassium concentration of >5.0 mmol/L were excluded.¹⁰

In EMPHASIS-HF, 2737 ambulatory patients with a LVEF $\leq$ 35% and New York Heart Association class II, treated with an ACE inhibitor or angiotensin receptor blocker and a β -blocker (unless contraindicated), were randomly assigned to eplerenone or matching placebo. Patients with a serum potassium level >5.0 mmol/L or an estimated glomerular filtration rate (eGFR) <30 mL/min per 1.73m^2 were excluded.¹¹

In TOPCAT, 3445 ambulatory patients with a LVEF $>45\%$ and either a history of HF hospitalization within the previous 12 months or elevated natriuretic peptide levels were randomly assigned to spironolactone or matching placebo. Patients with a serum potassium level >5.0 mmol/L, serum creatinine level >2.5 mg/dL, or an eGFR <25 mL/min per 1.73m^2 were excluded.⁹

In FINEARTS-HF, 6001 ambulatory and hospitalized patients with a LVEF $>40\%$, evidence of structural heart disease, New York Heart Association class II to IV, and elevated natriuretic peptide levels were randomly assigned to finerenone or matching placebo. Patients with a serum potassium level >5.0 mmol/L or an eGFR <25 mL/min per 1.73m^2 were excluded.¹²

All trials were assessed as high quality, but the risk of bias for TOPCAT with regard to the assessment of deviation from the intended intervention was high (Figure S2). As stated below, a sensitivity analysis only including the participants from the Americas was therefore undertaken to examine the impact of this potential bias (due to concerns about enrollment of patients without HF and nonadherence to trial treatment in the other participating countries).

Race

Patients were assigned a race subgroup on the case report form based on their self-identification. The prespecified race subgroups were as follows: "White," "Black," "Oriental," or "Other" (specify "Other") in RALES; "White," "Black," "Asian," "Other" (specify "Other") in EMPHASIS-HF; "White or White," "Black," "Asian," "American Indian or Alaska Native," "Native Hawaiian or Other Pacific Islander," or "Other" (specify "Other") in TOPCAT; and "White," "Black," "Asian," "American Indian or Alaska Native," "Native Hawaiian or Other Pacific Islander" or "not reported."

In the main analysis, race was categorized as Black and non-Black. Additional comparisons included (1) Black, White, and Asian participants; (2) Black and non-Black participants enrolled in the Americas only (where 91.3% of Black patients in these trials were recruited); and (3) Black and White participants enrolled in the Americas only.

Trial Outcomes

In the present analysis, the primary outcome was a composite of cardiovascular death or first HF hospitalization. Secondary outcomes examined were first HF hospitalization, total (first and recurrent) HF hospitalizations, cardiovascular death and total HF hospitalizations, cardiovascular death, and all-cause

death. In each trial, outcomes were adjudicated by a masked clinical end points committee. Cardiovascular death was analyzed according to the original definition reported in each of the trials.^{9–12} Data on urgent visits for worsening HF were included in FINEARTS-HF, and these were included in the definition of HF hospitalizations as a hospitalization.¹²

Changes in systolic blood pressure (SBP) and eGFR, calculated using the 2021 Chronic Kidney Disease Epidemiology Collaboration creatinine equation (which does not incorporate race),¹³ were also examined. In all trials, SBP and creatinine were measured at baseline. During follow-up, these measures were collected at varying time points across trials (Table S1). Therefore, analyses were restricted to time points that could be harmonized across studies. For SBP, data at baseline, 1 month, 9 months (8 months in TOPCAT, classified as 9 months), and 12 months (13 months in EMPHASIS-HF, classified as 12 months) were included. For eGFR, data at baseline, 6 months (5 months in EMPHASIS-HF and 4 months in TOPCAT, classified as 6 months), and 12 months (13 months in EMPHASIS-HF, classified as 12 months) were included.

Safety outcomes of interest included serum creatinine >2.5 and >3.0 mg/dL, a $>20\%$ and $>30\%$ decline in eGFR, serum potassium <3.5 , >5.5 , and >6 mmol/L, and SBP <90 and <100 mm Hg.

All efficacy analyses were performed according to the intention-to-treat principle. Safety analyses were only performed in patients who had received at least one dose of either MRA or placebo.

Statistical Analyses

Baseline characteristics were summarized as frequencies with percentages, means with standard deviations, or medians with interquartile ranges. Differences in baseline characteristics were tested using the χ^2 test for binary or categorical variables and the Kruskal-Wallis test and ANOVA test for non-normal and normally distributed continuous variables, respectively.

The incidence rates of clinical outcomes according to race were calculated as the number of events per 100 person-years, and 95% CI were estimated with Poisson regression models with robust SEs. Rate differences according to treatment assignment in Black and non-Black patients were estimated using differences in predicted event rates from Poisson models.

The effects of MRA versus placebo on clinical outcomes according to race were evaluated with Cox proportional hazards models for time-to-event data and negative binomial models for total events, and these models were stratified/adjusted by trial as appropriate. Hazard ratios (HRs) and rate ratios, respectively, with 95% CI were reported. No further adjustment for baseline characteristics was performed.

The change in SBP and eGFR from baseline to 12 months was analyzed using mixed-effect models for repeated measurements, adjusted for baseline value, visit, treatment-by-visit interaction, and trial. The least-squares mean differences with 95% CI were reported.

The effects of MRA versus placebo on safety events according to race were evaluated with logistic regression models, adjusted for trial. Odds ratios with 95% CI were reported.

The proportional hazards assumption was examined with scaled Schoenfeld residuals and log(−log[survival]) curves

and was not violated for any of the Cox proportional hazards or semiparametric proportional-rates models. The Wald test was used to test for interaction between the relative treatment effect of MRA and race for all efficacy end points, and the respective models included treatment assignment, race, and their interaction as covariates, in addition to those described above. Interaction for rate differences was tested in a linear model of rates, weighted by observed follow-up time, which included treatment assignment, race, and their interaction as covariates. For the safety outcomes, the Wald test was used to test for interaction between the treatment effect of MRA and race in a logistic regression model, which included treatment assignment, race, and their interaction as covariates.

All analyses were conducted using SAS version 9.4 (SAS Institute, Cary, NC) and STATA version 17.0 (College Station, TX).

RESULTS

Of the 13846 patients randomized in RALES, EMPHASIS-HF, TOPCAT, and FINEARTS-HF, 11505 (83.1%) identified as White, 577 (4.2%) as Black, 1363 (9.8%) as Asian, and 401 (2.9%) as other (ie, "American Indian or Alaska Native," "Native Hawaiian or Other Pacific Islander," "other," or "not reported"). The proportion of Black patients (ie, Black) in the 4 included trials was 7.2%, 2.4%, 8.8%, and 1.5%, respectively.

The median follow-up was 30 months overall, 25 months for Black patients, and 30 months for non-Black patients.

Patient Characteristics

Compared with non-Black patients, Black patients were younger and more often women, and they had higher heart rate and body mass index (Table 1). Black patients had higher eGFR than non-Black patients, assessed by the 2009 Chronic Kidney Disease Epidemiology Collaboration creatinine equation, but they had a significantly lower eGFR, assessed by the 2021 Chronic Kidney Disease Epidemiology Collaboration creatinine equation. Compared with non-Black patients, Black patients more often had a history of type 2 diabetes, but less often had a history of atrial fibrillation and myocardial infarction. Although Black and non-Black patients had similar NT-proBNP (N-terminal pro-B-type natriuretic peptide) levels, Black patients had worse New York Heart Association class and higher LVEF, and they were more likely to have a prior HF hospitalization (Table 1).

Regarding pharmacological therapy, Black patients were more frequently treated with a diuretic and digoxin and less often with a β -blocker than non-Black patients. A similar proportion of patients were treated with an ACE inhibitors/angiotensin receptor blocker and angiotensin receptor–neprilysin inhibitor (Table 1).

Effects of MRA on Clinical Outcomes According to Race

Overall, Black patients had a higher risk of all clinical outcomes than non-Black patients (Table 2; Figure 1). For the primary composite outcome of cardiovascular death or first HF hospitalization, the event rate in the placebo arm was 21.7 (95% CI, 17.6–26.8) per 100 person-years in Black patients and 12.2 (95% CI, 11.6–12.8) per 100 person-years in non-Black patients. The MRA versus placebo HR was 0.87 (95% CI, 0.66–1.15) in Black patients and 0.77 (95% CI, 0.72–0.82) in non-Black patients ($P_{\text{interaction}}=0.34$), with 3.9 and 2.7 fewer events per 100 person-years of treatment, respectively (Table 2; Figure 1). For a first HF hospitalization, the HR was 0.86 (95% CI, 0.63–1.17) in Black patients and 0.73 (95% CI, 0.68–0.80) in non-Black patients ($P_{\text{interaction}}=0.36$). The corresponding HRs for cardiovascular death were 0.75 (95% CI, 0.48–1.17) and 0.81 (95% CI, 0.74–0.90), respectively ($P_{\text{interaction}}=0.80$). For all-cause death, the HR was 0.95 (95% CI, 0.67–1.35) in Black patients and 0.84 (95% CI, 0.78–0.91) in non-Black patients ($P_{\text{interaction}}=0.46$; Table 2).

For cardiovascular death and total HF hospitalizations, the event rate in the placebo arm was 35.1 (95% CI, 28.0–43.8) per 100 person-years in Black patients and 19.6 (95% CI, 18.5–20.8) per 100 person-years in non-Black patients. The rate ratio was 0.78 (95% CI, 0.54–1.12) in Black patients and 0.72 (95% CI, 0.66–0.80) in non-Black patients ($P_{\text{interaction}}=0.85$), with 5.3 and 4.9 fewer HF hospitalizations per 100 person-years of treatment, respectively (Table 2). For total HF hospitalizations, the rate ratio was 0.83 (95% CI, 0.56–1.23) in Black patients and 0.68 (0.61–0.76) in non-Black patients ($P_{\text{interaction}}=0.58$).

The effects of MRA on clinical outcomes in patients with HFrEF and HFpEF, individually, were not modified by race (Table S2).

Effects of MRA on SBP and eGFR According to Race

Compared with placebo, MRA significantly reduced the mean SBP and eGFR from baseline to 12 months. The placebo-corrected mean change in SBP was greater in Black patients (−3.43 [95% CI, −6.76 to −0.09] mmHg) than in non-Black patients (−2.79 [−3.34 to −2.25] mmHg; $P_{\text{interaction}}=0.02$; Table 3; Figure 2). The placebo-corrected mean change in eGFR was −4.18 mL/min per 1.73m² (95% CI, −7.04 to −1.31 mL/min per 1.73m²) in Black patients and 3.38 mL/min per 1.73 m² (−3.86 to −2.90) in non-Black patients, with no interaction between race and effect of treatment ($P_{\text{interaction}}=0.39$; Table 3; Figure 2).

The effects of MRA on SBP and eGFR in patients with HFrEF and HFpEF, individually, were not modified by race, with one exception: MRA treatment reduced SBP to a greater extent in Black versus non-Black individuals with HFpEF ($P_{\text{interaction}}=0.003$; Table 3).

Table 1. Baseline Characteristics of Black and Non-Black Participants

	Non-Black patients (n=13 269)	Black patients (n=577)	P value
Demographics			
Age, mean (SD), y	69.9±9.7	64.1±11.5	<0.001
Sex, n (%)			<0.001
Men	8027 (60.5)	256 (44.4)	
Women	5242 (39.5)	321 (55.6)	
Race, n (%)			<0.001
White	11 505 (86.7)	0 (0.0)	
Black	0 (0.0)	577 (100.0)	
Asian	1363 (10.3)	0 (0.0)	
Other	401 (3.0)	0 (0.0)	
Geographic region, n (%)			<0.001
North America	1959 (14.8)	351 (60.8)	
Latin America	1286 (9.7)	176 (30.5)	
Western Europe	3260 (24.6)	15 (2.6)	
Central Europe	5296 (39.9)	0 (0.0)	
Asia-Pacific	1468 (11.1)	35 (6.1)	
Physiological measures			
Systolic blood pressure, mean (SD), mm Hg	127.4±16.1	128.2±18.7	0.27
Diastolic blood pressure, mean (SD), mm Hg	75.3±10.5	75.1±11.8	0.70
Heart rate, mean (SD), bpm	72.0±12.3	73.4±13.9	0.006
Body mass index, mean (SD), kg/m ²	29.8±6.2	35.1±9.1	<0.001
Body mass index, n (%), kg/m ²			<0.001
<30	6665 (57.0)	147 (32.6)	
≥30	5029 (43.0)	304 (67.4)	
Body weight, mean (SD), kg	81.8±19.4	90.9±28.0	<0.001
LVEF, %, mean (SD)	45.3±15.0	46.9±16.8	0.016
LVEF, %, n (%)			0.85
<40	4352 (33.2)	192 (33.6)	
>40	8765 (66.8)	380 (66.4)	
NYHA class, n (%)			<0.001
I, II	8876 (66.9)	310 (54.0)	
III, IV	4385 (33.1)	264 (46.0)	
Current smoker, n (%)	1112 (9.5)	52 (11.4)	0.17
Biomarkers			
NT-proBNP, median (IQR), pg/mL	1013.5 (450.9–1928.3)	925.6 (418.1–1943.8)	0.55
Creatinine, mean (SD), μmol/L	1.1±0.3	1.2±0.4	<0.001
eGFR, mL/min per 1.73 m ²			
2009 CKD-EPI creatinine equation: mean (SD)	63.6±19.1	68.9±23.9	<0.001
2009 CKD-EPI creatinine equation: n (%)			0.015
<60	6008 (45.3)	232 (40.2)	
≥60	7242 (54.7)	345 (59.8)	
2021 CKD-EPI creatinine equation: mean (SD)	67.4±19.7	62.9±21.2	<0.001
2021 CKD-EPI creatinine equation: n (%)			<0.001
<60	5048 (38.1)	281 (48.7)	
≥60	8202 (61.9)	296 (51.3)	
Potassium, mean (SD), mmol/L	4.3±0.5	4.2±0.5	<0.001
Sodium, mean (SD), mmol/L	140.5±3.7	139.9±3.4	<0.001

(Continued)

Table 1. Continued

	Non-Black patients (n=13 269)	Black patients (n=577)	P value
Medical history, n (%)			
Hospitalization for HF	7237 (61.7)	309 (67.6)	0.011
Type 2 diabetes	4568 (34.4)	232 (40.2)	0.004
Hypertension	10 247 (77.2)	435 (75.4)	0.30
Atrial fibrillation	5406 (40.8)	108 (18.7)	<0.001
Myocardial infarction	4212 (31.8)	74 (12.8)	<0.001
Stroke	1194 (10.2)	41 (9.0)	0.40
Treatment, n (%)			
ACE inhibitors/ARB	10 819 (81.6)	474 (82.3)	0.70
ARNI	503 (8.5)	10 (11.4)	0.34
β-Blocker	9942 (75.0)	374 (64.9)	<0.001
Any diuretic	12 037 (90.8)	538 (93.4)	0.036
Digoxin	2616 (19.7)	153 (26.6)	<0.001

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration; eGFR, estimated glomerular filtration rate; HF, heart failure; IQR, interquartile range; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; and NYHA, New York Heart Association.

Effects of MRA on Safety Events According to Race

Compared with participants in the placebo arm, those in the MRA arm were more likely to experience hypotension, hyperkalemia, elevated creatinine levels, and reductions in eGFR. Conversely, MRA treatment reduced the risk of hypokalemia compared with placebo. Overall, these effects were not modified by race (Table 4).

The effects of MRA on safety outcomes in patients with HFrEF and HFpEF, individually, were not modified by race, with a few exceptions: MRA increased the odds of experiencing an increase in serum creatinine >2.5 mg/dL in non-Black participants, but not in Black patients, with HFrEF ($P_{\text{interaction}}=0.03$). MRA increased the odds of experiencing a decrease in SBP<100 mmHg to a greater extent in Black versus non-Black individuals with HFpEF ($P_{\text{interaction}}=0.004$; Table S3).

Black, White, and Asian Participants

Overall, race (ie, Black, White, and Asian race) did not modify the effects of MRA versus placebo on any of the outcomes with one exception: MRA treatment reduced SBP to a greater extent in Black and Asian versus non-Black individuals (Tables S4 through S7).

Black and Non-Black Participants Enrolled in the Americas Only

Overall, race (ie, Black and non-Black race) did not modify the effects of MRA versus placebo on any of the outcomes among participants enrolled in the Americas (Tables S8 through S11).

Black and White Participants Enrolled in the Americas Only

Overall, race (ie, Black and White race) did not modify the effects of MRA versus placebo on any of the outcomes among participants enrolled in the Americas (Tables S12 through S15).

DISCUSSION

In this patient-level pooled meta-analysis of almost 14 000 participants from large placebo-controlled clinical trials, MRAs reduced the absolute and relative risk of HF hospitalization (both first and total HF hospitalizations), cardiovascular death, and all-cause death with no evidence of heterogeneity in treatment effect between Black and non-Black patients. The effects of MRAs on experiencing hypotension, hyperkalemia, hypokalemia, or elevated creatinine levels did not differ meaningfully by race. The benefits of MRAs in HFrEF and HFpEF, individually, were consistent irrespective of race.

Consistent with previous reports, participants who self-identified as Black had a higher prevalence of HF and higher event rates than non-Black participants in the present analysis.^{14–16} However, the findings from our, and previous, analyses should be interpreted with caution. Race is recognized to be a social, rather than a biological construct, and we do not have data on social determinants of health, which are increasingly being recognized as important factors for the observed disparities in outcomes between Black patients and others.^{14–16} For example, availability of and access to care in the community as opposed to a hospital may differ according to



Table 2. Effects of MRAs Compared With Placebo on Efficacy Outcomes in Black and Non-Black Participants

	Non-Black patients (n=13269)		Black patients (n=577)		P value for interaction
	MRA (n=6616)	Placebo (n=6653)	MRA (n=295)	Placebo (n=282)	
Cardiovascular death or first HF hospitalization*†					
n (%)	1442 (21.8)	1788 (26.9)	97 (32.9)	106 (37.6)	
Event rate per 100 PY (95% CI)	9.5 (9.0 to 10.0)	12.2 (11.6 to 12.8)	17.7 (14.3 to 21.9)	21.7 (17.6 to 26.8)	
Rate difference (95% CI)	2.7 (1.9 to 3.5)		3.9 (−2.0 to 9.9)		0.69
HR (95% CI)‡	0.77 (0.72 to 0.82)		0.87 (0.66 to 1.15)		0.34
First HF hospitalization*					
n (%)	989 (14.9)	1286 (19.3)	76 (25.8)	85 (30.1)	
Event rate per 100 PY (95% CI)	6.5 (6.1 to 6.9)	8.8 (8.3 to 9.3)	13.7 (10.8 to 17.5)	17.1 (13.6 to 21.6)	
Rate difference (95% CI)	2.3 (1.6 to 2.9)		3.4 (−1.8 to 8.6)		0.67
HR (95% CI)‡	0.73 (0.68 to 0.80)		0.86 (0.63 to 1.17)		0.36
Cardiovascular death*†					
n (%)	741 (11.2)	891 (13.4)	34 (11.5)	44 (15.6)	
Event rate per 100 PY (95% CI)	4.5 (4.2 to 4.9)	5.5 (5.1 to 5.9)	5.1 (3.7 to 7.1)	7.1 (5.2 to 9.6)	
Rate difference (95% CI)	1.0 (0.5 to 1.5)		2.0 (−0.8 to 4.8)		0.48
HR (95% CI)‡	0.81 (0.74 to 0.90)		0.75 (0.48 to 1.17)		0.80
Total HF hospitalizations*					
n (%)	1791	2461	171	183	
Event rate per 100 PY (95% CI)	11.0 (10.1 to 11.8)	15.3 (14.3 to 16.3)	25.7 (18.9 to 34.8)	29.6 (23.0 to 38.0)	
Rate difference (95% CI)	4.3 (3.0 to 5.6)		3.9 (−7.0 to 14.8)		0.94
RR (95% CI)§	0.68 (0.61 to 0.76)		0.83 (0.56 to 1.23)		0.58
Cardiovascular death and total HF hospitalizations*†					
n (%)	2405	3167	198	217	
Event rate per 100 PY (95% CI)	14.7 (13.8 to 15.7)	19.6 (18.5 to 20.8)	29.7 (22.8 to 38.8)	35.1 (28.0 to 43.8)	
Rate difference (95% CI)	4.9 (3.4 to 6.4)		5.3 (−5.9 to 16.6)		0.94
RR (95% CI)§	0.72 (0.66 to 0.80)		0.78 (0.54 to 1.12)		0.85
All-cause death					
n (%)	1142 (17.3)	1332 (20.0)	61 (20.7)	62 (22.0)	
Event rate per 100 PY (95% CI)	6.9 (6.5 to 7.4)	8.2 (7.8 to 8.6)	9.2 (7.1 to 11.7)	10.0 (7.7 to 12.9)	
Rate difference (95% CI)	1.2 (0.6 to 1.8)		0.8 (−2.6 to 4.3)		0.81
HR (95% CI)‡	0.84 (0.78 to 0.91)		0.95 (0.67 to 1.35)		0.46

HF indicates heart failure; HR, hazard ratio; MRA, mineralocorticoid-receptor antagonist; PY, person-years; and RR, rate ratio.

*Urgent visits for worsening HF were included in the definition of HF hospitalizations as a hospitalization equivalent.

†The definition of cardiovascular death was that used in the original trials.

‡HRs were estimated using Cox proportional hazards regression models stratified by trial.

§Rate ratios were estimated using negative binomial models adjusted for trial.

socioeconomic circumstances, including neighborhood and other variables.

Differences in renin-angiotensin-aldosterone system activity have been consistently reported in Black compared with White people.^{1–4} Although Black patients may have less blood pressure reduction with renin-angiotensin system inhibitors (ie, ACE inhibitors and angiotensin receptor blockers) than non-Black patients, a recent meta-analysis of all the large placebo-controlled outcome trials with renin-angiotensin system inhibitors in HFrEF demonstrated similar absolute benefits from these drugs in Black and non-Black participants.¹⁷ However, the evidence regarding the efficacy of MRAs across the spectrum of LVEF according to race had been conflicting. A

post hoc analysis of participants with HFrEF enrolled in RALES suggested that race modified the effect of spironolactone on the composite of all-cause death or first HF hospitalization, with a greater benefit of spironolactone in non-Black compared with Black participants ($P_{\text{interaction}}=0.032$).⁷ A similar trend was observed for the primary outcome of all-cause death, although race did not significantly modify the overall survival benefit with spironolactone.⁷ In contrast, in a post hoc analysis of participants with HF with mildly reduced ejection fraction/HFpEF enrolled in TOPCAT in the Americas, race did not significantly modify the effects of spironolactone on the primary composite end point, cardiovascular death, or first HF hospitalization ($P_{\text{interaction}}>0.19$ for all outcomes).⁸

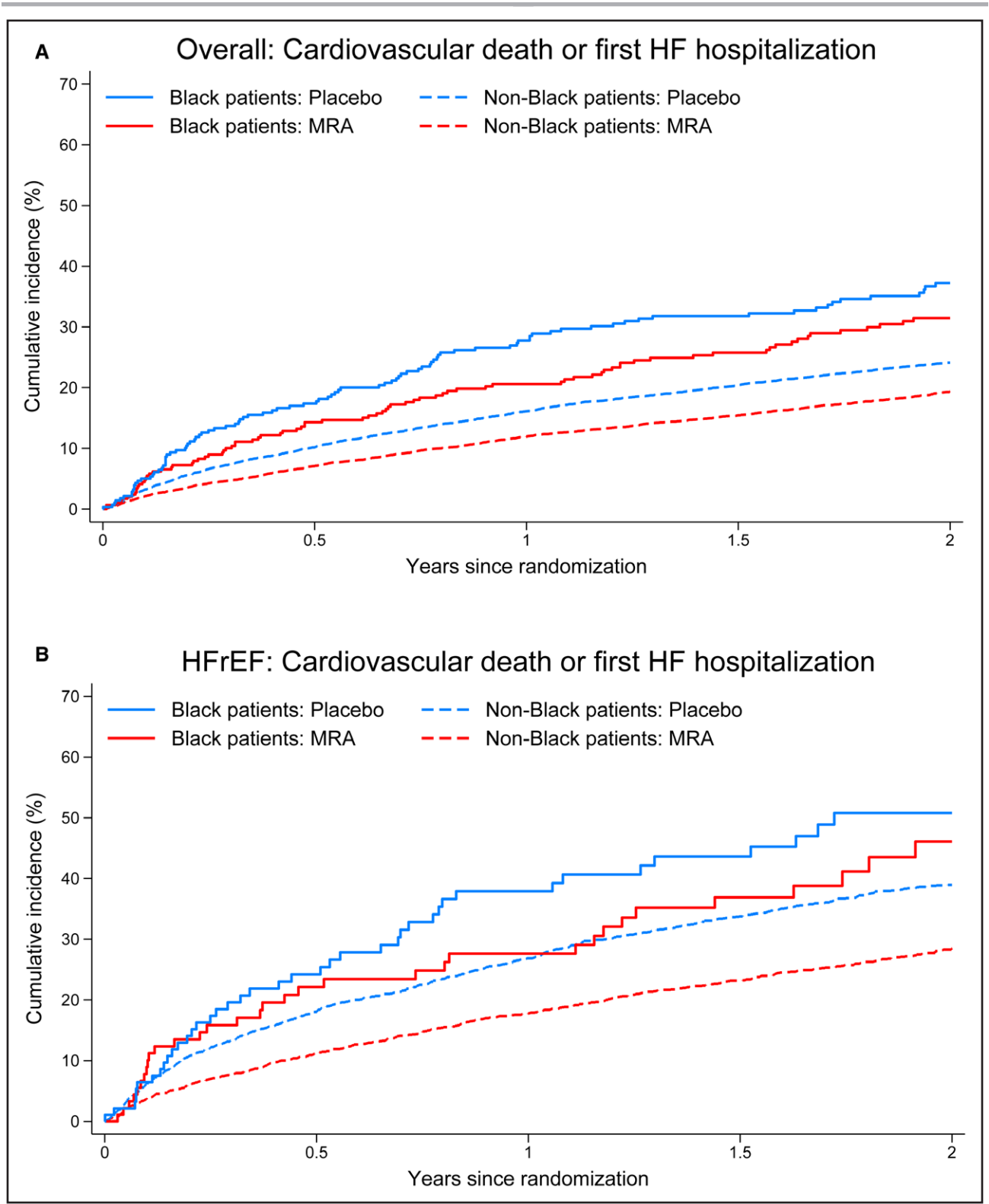


Figure 1. Cumulative incidence of cardiovascular death or first heart failure (HF) hospitalization according to treatment assignment and race.

A, Overall. **B**, Heart failure with reduced ejection fraction (HFrEF). **C**, Heart failure with preserved ejection fraction (HFpEF). MRA, mineralocorticoid-receptor antagonist. (Continued)

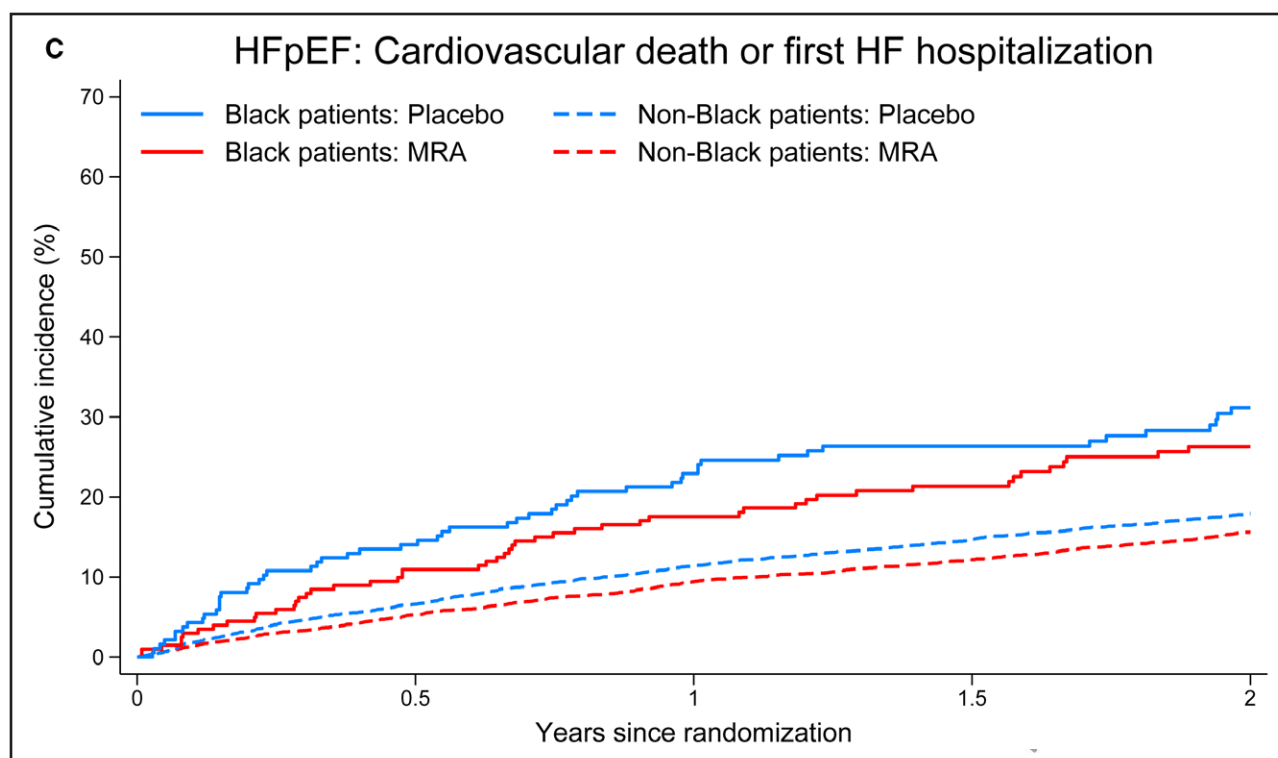


Figure 1 Continued.

However, TOPCAT did not show a significant benefit of spironolactone overall.⁹ The effects of eplerenone have not been reported by race among participants with HFrEF enrolled in EMPHASIS-HF. More recently, in FINEARTS-HF, the benefit of finerenone on cardiovascular death and total HF events in HF with mildly reduced ejection fraction/HFpEF was consistent across racial groups.¹² Unfortunately, Black individuals are underrepresented in clinical trials of HF, and the few events in a small number of Black participants in individual trials makes it difficult to obtain a robust estimate of the effect of a therapy and to ensure generalizability of the results to this high-risk population.

The present patient-level pooled meta-analysis of the 4 large placebo-controlled MRA trials, including 577 Black patients and 13 269 non-Black patients, extends previous findings by providing more reliable and precise estimates of the effect of MRAs in Black patients and by demonstrating consistent clinical benefit of treatment on key clinical outcomes regardless of race. Specifically, we found that the benefits of MRAs on clinical outcomes were not modified by race, with consistent absolute and relative reductions in HF hospitalization (both first and total HF hospitalizations), cardiovascular death, and all-cause death, in Black and non-Black patients. Although modest heterogeneity in treatment effect cannot be excluded given the limited number of Black participants and clinical events, our findings do not suggest a large clinically meaningful difference in response to MRAs between Black and non-Black patients.

Overall, the benefits of MRAs on clinical outcomes in patients with HFrEF and HFpEF, individually, were not modified by race. Although a numerical trend towards a lesser benefit of MRAs on first HF hospitalization was observed in Black versus non-Black patients with HFrEF, reassuringly, this was not the case for the composite of cardiovascular death and total HF hospitalizations and each of its components. However, it should be noted that MRA treatment did not reduce cardiovascular death (or all-cause death) in HFpEF overall (or by race group), in contrast to HFrEF.

Compared with placebo, MRAs reduced the mean SBP from baseline to 12 months. However, there was a nominally significant interaction between race and the treatment effect on SBP, such that MRAs appeared to reduce the mean SBP to a greater extent in Black patients. This contrasts with what has been shown for ACE inhibitors and angiotensin receptor blockers, including in HFrEF.¹⁷ However, given that the interaction tests were not adjusted for multiple comparisons, this borderline interaction could constitute a chance finding. Decreases in eGFR were more common with MRA treatment, and there was no evidence of effect modification in the magnitude of this effect by race.

Although serious adverse events were uncommon overall, hypotension, hyperkalemia, and elevated creatinine levels occurred more frequently with MRA treatment than with placebo. Conversely, the risk of hypokalemia was halved by MRA treatment. Reassuringly, these effects were not modified by race.

Table 3. Change in Systolic Blood Pressure and Estimated Glomerular Filtration Rate With MRAs Compared With Placebo From Baseline to 12 Months in Black and Non-Black Participants

	Non-Black patients		Black patients		P value for interaction
	MRA	Placebo	MRA	Placebo	
Overall					
Systolic blood pressure, mmHg					0.02
Mean change (95% CI)	−2.72 (−3.10 to −2.33)	0.07 (−0.31 to 0.46)	−1.61 (−3.90 to 0.68)	1.82 (−0.60 to 4.23)	
Placebo-corrected mean change (95% CI)	−2.79 (−3.34 to −2.25)		−3.43 (−6.76 to −0.09)		
eGFR, mL/min per 1.73m ²					0.39
Mean change (95% CI)	−3.85 (−4.19 to −3.51)	−0.47 (−0.81 to −0.13)	−4.80 (−6.77 to −2.82)	−0.62 (−2.69 to 1.44)	
Placebo-corrected mean change (95% CI)	−3.38 (−3.86 to −2.90)		−4.18 (−7.04 to −1.31)		
HFrEF					
Systolic blood pressure, mmHg					0.25
Mean change (95% CI)	−1.47 (−2.24 to −0.69)	1.22 (0.42 to 2.01)	−3.25 (−7.66 to 1.17)	0.81 (−3.67 to 5.29)	
Placebo-corrected mean change (95% CI)	−2.68 (−3.79 to −1.58)		−4.06 (−10.35 to 2.24)		
eGFR, mL/min per 1.73m ²					0.45
Mean change (95% CI)	−3.77 (−4.47 to −3.08)	−1.25 (−1.96 to −0.54)	0.36 (−4.33 to 5.05)	1.40 (−3.29 to 6.09)	
Placebo-corrected mean change (95% CI)	−2.52 (−3.52 to −1.53)		−1.04 (−7.69 to 5.61)		
					
HFpEF					
Systolic blood pressure, mmHg					0.03
Mean change (95% CI)	−3.21 (−3.65 to −2.77)	−0.33 (−0.77 to 0.11)	−1.23 (−3.90 to 1.45)	2.21 (−0.65 to 5.07)	
Placebo-corrected mean change (95% CI)	−2.88 (−3.50 to −2.26)		−3.44 (−7.36 to 0.48)		
eGFR, mL/min per 1.73 m ²					0.70
Mean change (95% CI)	−3.88 (−4.27 to −3.49)	−0.19 (−0.58 to 0.20)	−6.73 (−8.80 to −4.65)	−1.49 (−3.70 to 0.73)	
Placebo-corrected mean change (95% CI)	−3.69 (−4.23 to −3.14)		−5.24 (−8.28 to −2.20)		

eGFR is calculated according to the 2021 Chronic Kidney Disease Epidemiology Collaboration equation. eGFR indicates estimated glomerular filtration rate; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; and MRA, mineralocorticoid-receptor antagonist.

Collectively, these data from a large patient-level pooled meta-analysis underscore the substantial and clinically meaningful benefits and tolerability of MRAs in both Black and non-Black patients with HF across the spectrum of LVEF and argue against the use of race to guide prescribing thresholds, dose selection, or monitoring strategies for MRAs in patients with HF. However, despite the well-established and substantial benefits of MRAs and other disease-modifying HF therapies, Black patients with HF are less likely to receive certain established guideline-directed medical therapy with proven benefit than others with this condition. Failure to prescribe therapies with proven efficacy in high-risk populations will only serve to exacerbate preexisting healthcare disparities, and efforts to ensure more equitable use of MRAs may be needed to improve outcomes in Black patients with HF.^{18,19}

Limitations

The findings of this study should be viewed in the context of potential limitations. First, patients enrolled in clinical

trials are selected according to specific inclusion and exclusion criteria, and our results may not be generalizable to all patients with HF in the general population. Second, although we found no heterogeneity in the effect of MRAs in Black patients, compared with other participants, subgroup analyses like these should be interpreted carefully because the interaction tests were not sufficiently powered to detect modest differences between groups. Accordingly, this study primarily examines the consistency of the effects of MRAs in HF across racial groups. Third, the relative representation of Black patients in the trials included in this meta-analysis was modest, reflecting their underrepresentation in clinical trials. Adequate inclusion of Black patients (and other underrepresented groups) is necessary to ensure the generalizability of study findings to all patients with HF. Fourth, data on socioeconomic status and other social determinants of health were not collected, and the social and structural factors that may underlie any race-related differences in outcomes could not be fully examined. Fifth, the use of self-identified race as a dichotomous variable does not

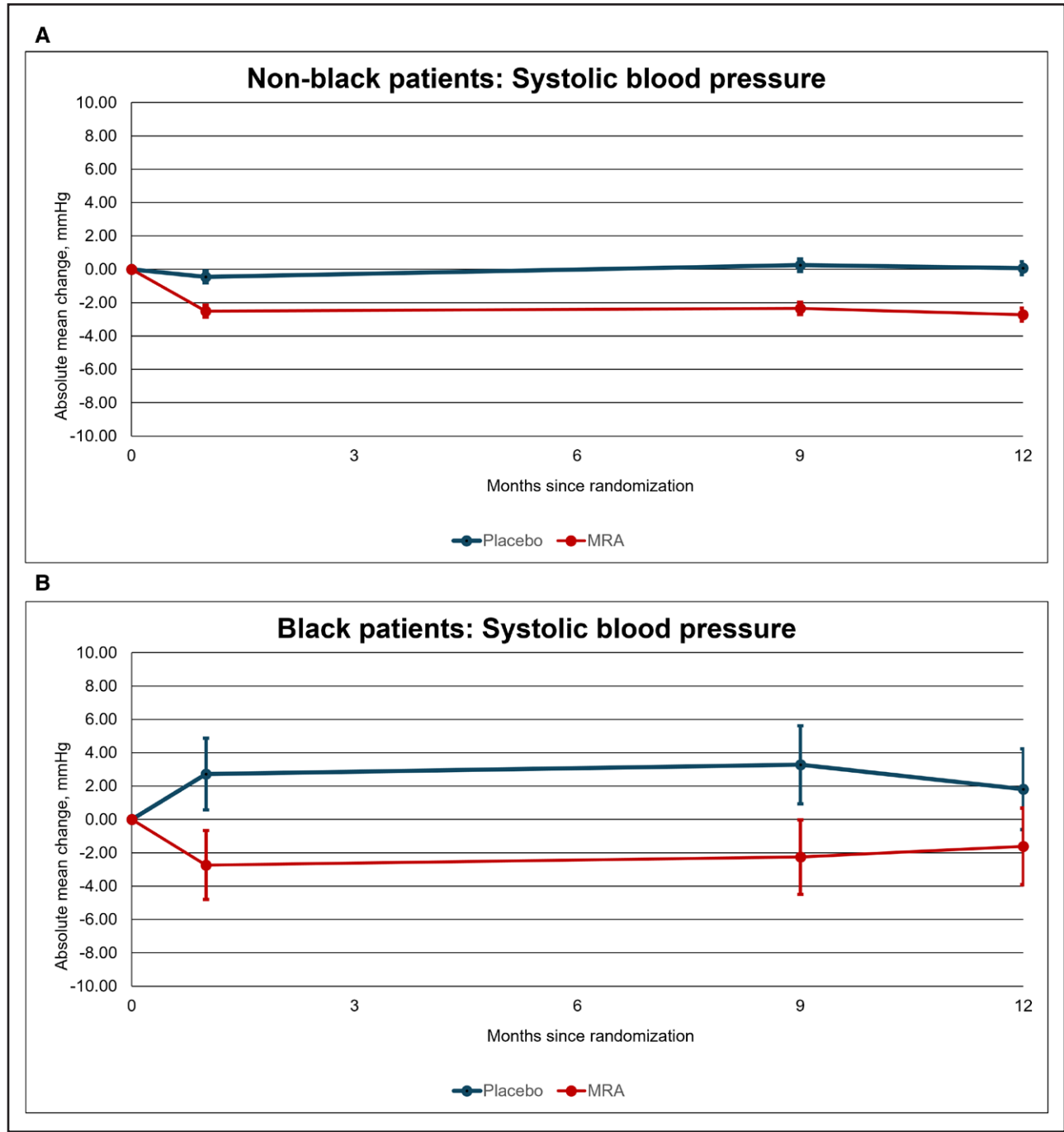


Figure 2. Change in systolic blood pressure and estimated glomerular filtration rate with mineralocorticoid-receptor antagonists (MRAs) compared with placebo from baseline to 12 months.
A, Non-Black patients: Systolic blood pressure. **B**, Black patients: Systolic blood pressure. **C**, Non-Black patients: Estimated glomerular filtration rate. **D**, Black patients: Estimated glomerular filtration rate. In RALES (Randomized Aldactone Evaluation Study), systolic blood pressure was measured after 1 month, 9 months, and 12 months. In EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure), systolic blood pressure was measured after 1 month, 9 months, and 13 months (classified as 12 months). In TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist), systolic blood pressure was measured after 1 month, 8 months (classified as 9 months), and 12 months. In FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure), systolic blood pressure was measured after 1, 9, and 12 months. Estimated glomerular filtration rate is calculated according to the 2021 Chronic Kidney Disease Epidemiology Collaboration equation. In RALES, creatinine was measured after 6 and 12 months. In EMPHASIS-HF, creatinine was measured after 5 months (classified as 6 months) and 13 months (classified as 12 months). In TOPCAT, creatinine was measured after 4 months (classified as 6 months) and 12 months. In FINEARTS-HF, creatinine was measured after 6 and 12 months. (Continued)

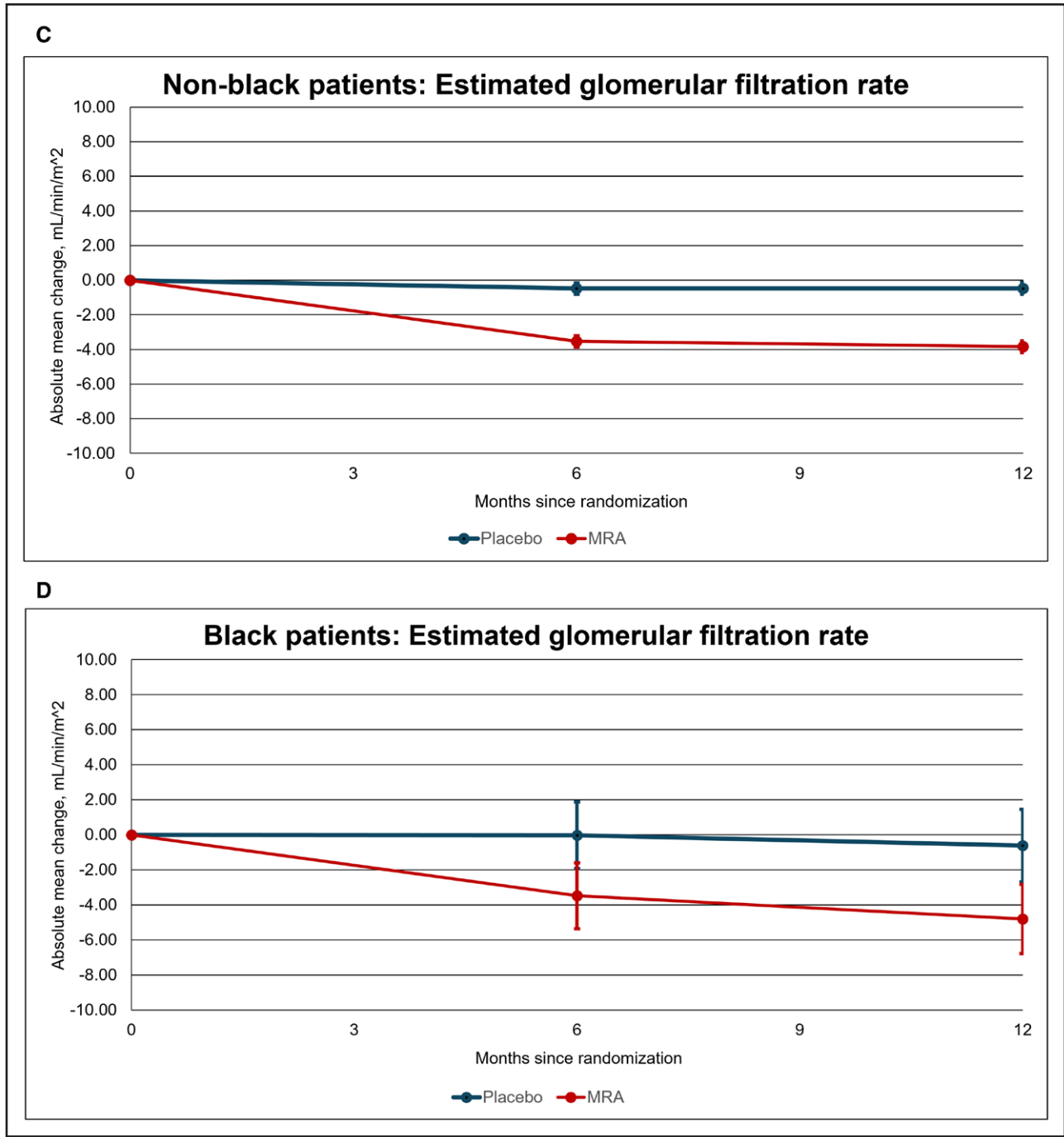


Figure 2 Continued.

capture the heterogeneity within racial groups, including potential genetic, biological, environmental, and sociocultural factors that may influence outcomes and treatment response. Sixth, the trials included in this meta-analysis were conducted over a 25-year period, during which the management of HF evolved substantially. Finally, the trials included in this meta-analysis evaluated different generations of MRAs across distinct HF phenotypes and eras of background therapy. Although spironolactone, eplerenone, and finerenone differ in structure, selectivity,

and affinity, they all act through antagonism of the mineralocorticoid receptor. Therefore, while differences in the magnitude of treatment benefit across individual trials, agents, and HF phenotypes are biologically and clinically plausible, large differences in the direction of treatment effect according to race would be less expected. Nevertheless, this study was not designed to formally compare individual agents, HF phenotypes, or to establish a class effect, which would require head-to-head or specifically powered prospective trials.

Table 4. Effects of MRAs Compared With Placebo on Safety Outcomes in Black and Non-Black Participants

	Non-Black patients (n=13 246)		Black patients (n=577)		P value for interaction
	MRA (n=6602)	Placebo (n=6644)	MRA (n=295)	Placebo (n=282)	
Hypotension					
Systolic blood pressure <90 mmHg					0.51
n (%)	342/6461 (5.3)	232/6493 (3.6)	19/285 (6.7)	10/271 (3.7)	
OR (95% CI)*	1.52 (1.28–1.80)		1.89 (0.85–4.18)		
Systolic blood pressure <100 mmHg					0.13
n (%)	1228/6461 (19.0)	940/6493 (14.5)	71/285 (24.9)	41/271 (15.1)	
OR (95% CI)*	1.39 (1.27–1.53)		2.00 (1.29–3.09)		
Elevated serum creatinine					
≥2.5 mg/dL					0.18
n (%)	335/6293 (5.3)	203/6310 (3.2)	31/276 (11.2)	27/263 (10.3)	
OR (95% CI)*	1.70 (1.42–2.03)		1.10 (0.64–1.91)		
≥3.0 mg/dL					0.12
n (%)	146/6293 (2.3)	84/6310 (1.3)	12/276 (4.3)	13/263 (4.9)	
OR (95% CI)*	1.76 (1.35–2.31)		0.85 (0.38–1.91)		
Reduction in eGFR					
>20%					0.78
n (%)	3084/6049 (51.0)	2396/6098 (39.3)	154/260 (59.2)	122/252 (48.4)	
OR (95% CI)*	1.62 (1.51–1.75)		1.53 (1.07–2.18)		
>30%					0.62
n (%)	1876/6049 (31.0)	1258/6098 (20.6)	109/260 (41.9)	69/252 (27.4)	
OR (95% CI)*	1.74 (1.60–1.89)		1.90 (1.30–2.76)		
Elevated serum potassium					
>5.5 mmol/L					0.86
n (%)	878/6443 (13.6)	420/6468 (6.5)	31/284 (10.9)	13/269 (4.8)	
OR (95% CI)*	2.27 (2.01–2.57)		2.35 (1.20–4.61)		
>6.0 mmol/L					0.49
n (%)	186/6443 (2.9)	91/6468 (1.4)	10/284 (3.5)	3/269 (1.1)	
OR (95% CI)*	2.08 (1.62–2.69)		3.30 (0.89–12.14)		
Reduced serum potassium					
<3.5 mmol/L					0.82
n (%)	463/6443 (7.2)	863/6468 (13.3)	41/284 (14.4)	66/269 (24.5)	
OR (95% CI)*	0.50 (0.44–0.56)		0.53 (0.34–0.82)		

Safety outcomes were examined in patients who received at least one dose of randomly assigned treatment. eGFR indicates estimated glomerular filtration rate; MRA, mineralocorticoid-receptor antagonist; and OR, odds ratio.

*ORs were estimated using logistic regression models adjusted for trial.

Conclusions

In this patient-level pooled meta-analysis of almost 14 000 participants from 4 placebo-controlled clinical trials, MRAs reduced the absolute and relative risk of HF hospitalization (both first and total HF hospitalizations), cardiovascular death, and all-cause death with no evidence of heterogeneity in treatment effect between Black and non-Black patients with HF. The benefits of MRAs in HFrEF and HFpEF, individually, were consistent regardless of race. The effects of MRAs on experiencing adverse events did not differ meaningfully by race.

ARTICLE INFORMATION

Received February 10, 2026; accepted July 20, 2026.

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Sources of Funding

The RALES trial (Randomized Aldactone Evaluation Study) was supported by a grant from Searle Pharmaceuticals. The EMPHASIS-HF trial (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure) was sponsored by Pfizer. The TOPCAT trial (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) was supported by the National Heart, Lung, and Blood Institute. The FINEARTS-HF trial (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure) was sponsored by Bayer.

Disclosures

Dr Butt reports advisory board honoraria from Abbott, AstraZeneca, and Bayer; consultant honoraria from AstraZeneca and Bayer; travel grants from AstraZeneca, Bayer, and Boehringer Ingelheim. Dr Jhund reports speakers' fees from AstraZeneca, Novartis, Alkem Metabolics, ProAdWise Communications, Sun Pharmaceuticals; advisory board fees from AstraZeneca, Boehringer Ingelheim, Novartis; research funding from AstraZeneca, Boehringer Ingelheim, Analog Devices Inc, Roche Diagnostics. Dr Jhund's employer, the University of Glasgow, has been remunerated for clinical trial work from AstraZeneca, Bayer AG, Novartis and Novo Nordisk. Director, GCTP Ltd, Dr Vardeny has received grants from AstraZeneca and Cardior; and institutional research support from Bayer and Cardurion. Dr Claggett has received personal consulting fees from Alnylam, Bristol Myers Squibb (BMS), Cardior, Cardurion, Corvia, CVRx, Eli Lilly, Intellia, Rocket, and has served on a data safety monitoring board for Novo Nordisk. Dr Vaduganathan has received research grant support, served on advisory boards, or had speaker engagements with American Regent, Amgen, AstraZeneca, Bayer AG, Baxter Healthcare, BMS, Boehringer Ingelheim, Chiesi, Cytokinetics, Fresenius Medical Care, Idorsia Pharmaceuticals, Lexicon Pharmaceuticals, Merck, Milestone Pharmaceuticals, Novartis, Novo Nordisk, Pharmacosmos, Relaysa, Roche Diagnostics, Sanofi, and Tricog Health, and participates on clinical trial committees for studies sponsored by AstraZeneca, Galmed, Novartis, Bayer AG, Occlutech, and Impulse Dynamics. Dr Desai has received institutional research grants (to Brigham and Women's Hospital) from Abbott, Alnylam, AstraZeneca, Bayer, Novartis, and Pfizer as well as personal consulting fees from Abbott, Alnylam, AstraZeneca, Bayer, Biofourmis, Boston Scientific, Medpace, Medtronic, Merck, Novartis, Parexel, Porter Health, Regeneron, River2Renal, Roche, Veristat, Verily, Zyds. Dr Pfeffer received research grant support through his institution from Novartis and reports consulting fees from Alnylam, AstraZeneca, Boehringer Ingelheim and Eli Lilly Alliance, Corvidia, DalCor, GlaxoSmithKline, Lexicon, National Heart, Lung, and Blood Institute (NHLBI) CONNECTS (Master Protocol Committee), Novartis, Novo Nordisk, Peerbridge, and Sanofi. Dr Lam has received research support from Novo Nordisk and Roche Diagnostics; has received consulting fees from Alleviant Medical, Allysta Pharma, AnaCardio AB, Applied Therapeutics, AstraZeneca, Bayer, Biopeutics, Boehringer Ingelheim, Boston Scientific, BMS, CardioRenal, CPC Clinical Research, Eli Lilly, Impulse Dynamics, Intellia Therapeutics, Ionis Pharmaceutical, Janssen Research & Development LLC, Medscape/WebMD Global LLC, Merck, Novartis, Novo Nordisk, Prosciento Inc, Quidel Corporation, Radcliffe Group Ltd, Recardio Inc, ReCor Medical, Roche Diagnostics, Sanofi, Siemens Healthcare Diagnostics and Us2.ai; and is a cofounder & nonexecutive director of US2.AI. Dr Senni has served on Advisory Boards, Consultancy and Honoraria for Novartis, Abbott, Merck, MSD, Vifor, AstraZeneca, Cardurion, Novo Nordisk, Bayer, and Boehringer Ingelheim. Dr Shah has received research grants from National Institute for Health (NIH; U54 HL160273, X01 HL169712, R01 HL140731, and R01 HL149423), American Heart Association (24SFRNPCN1291224), AstraZeneca, Corvia, and Pfizer and consulting fees from Abbott, Alleviant, AstraZeneca, Amgen, Aria CV, Axon Therapies, Bayer, Boehringer Ingelheim, Boston Scientific, BMS, Cycleron, Cytokinetics, Edwards Lifesciences, Eidos, Imara, Impulse Dynamics, Intellia, Ionis, Lilly, Merck, MyoKardia, Novartis, Novo Nordisk, Pfizer, Prothena, Regeneron, Rivus, Sardocor, Shifamed, Tenax, Tenaya, and Ultromics. Dr Voors reports research grants from Boehringer Ingelheim and Roche Diagnostics; consulting fees from Merck, Bayer, Anacardio, AstraZeneca, Boehringer Ingelheim, BMS, Corteria, Cytokinetics, Eli Lilly, Moderna, Novartis, Novo Nordisk, Roche Diagnostics, and Tectonic. Dr Veldhuisen received fees for serving on a steering committee and travel support from ARCA Biopharma and Corvia Medical. Dr Verma has received speaker honoraria or consulting fees from Abbott, Amarin, AstraZeneca, Bayer, Boehringer Ingelheim, Canadian Medical, Surgical Knowledge Translation

Research Group, Eli Lilly, HLS Therapeutics, Janssen, Merck Sharp and Dohme, Novartis, Novo Nordisk, Pfizer, PhaseBio, and TIMI. Dr Lewis reports grants or contracts from Merck, DalCor, American Heart Association; consulting fees from Merck, AstraZeneca, Intellia, DalCor, Akebia; participation in a data safety monitoring board or advisory board for NHLBI; leadership or fiduciary role in American Heart Association. Dr Zannad reports personal fees from 89Bio, Abbott, Acceleron, Applied Therapeutics, Bayer, Betagenon, Boehringer, BMS, CVRx, Cambrian, Cardior, Cereno Pharmaceutical, Cellprothera, CEVA, Inventiva, KBP, Merck, Novo Nordisk, Owkin, Otsuka, Roche Diagnostics, Northsea, USa2, having stock options at G3Pharmaceutical and equities at Cereno, Cardiorenal, Eshmun Clinical Research, and being the founder of Cardiovascular Clinical Trials. Dr Pitt is a consultant for Bayer, Astra Zeneca, Boehringer Ingelheim, Lexicon, BMS, KBP Biosciences (stock/stock options), Sarfez Pharmaceuticals (stock/stock options), Pharmaceuticals (stock/stock options), SQinnovations (stock/stock options), G3 Pharmaceuticals, Sea Star medical (stock/stock options), Vifor (stock/stock options) Prointel (stock/stock options), Brainstorm Medical (stock/stock options). US Patent 9931412: Site-specific delivery of eplerenone to the myocardium, US Patent pending 63/045,783: Histone modulating agents for the prevention and treatment of organ failure. Dr Solomon has received research grants from Alexion, Alnylam, AstraZeneca, Bellerophon, Bayer, BMS, Boston Scientific, Cytokinetics, Edgewise, Eidos, Gossamer, GSK, Ionis, Lilly, MyoKardia, NIH/NHLBI, Novartis, NovoNordisk, Respicardia, Sanofi Pasteur, Theracos, US2.AI and has consulted for Abbott, Action, Akros, Alexion, Alnylam, Amgen, Arena, AstraZeneca, Bayer, Boehringer Ingelheim, BMS, Cardior, Cardurion, Corvia, Cytokinetics, Daiichi Sankyo, GSK, Lilly, Merck, Myokardia, Novartis, Roche, Theracos, Quantum Genomics, Janssen, Cardiac Dimensions, Tenaya, Sanofi Pasteur, Dinaqor, Trembeau, CellProThera, Moderna, American Regent, Sarepta, Lexicon, Anacardio, Akros, Valo. Dr McMurray reports payments through Glasgow University from work on clinical trials, consulting and grants from: Amgen, AstraZeneca, Bayer, Cardurion, Cytokinetics, GSK and Novartis, British Heart Foundation, NIH-NHLBI, Boehringer Ingelheim, SQ Innovations, Catalyze Group. Personal consultancy fees from: Alnylam Pharmaceuticals, Amgen, AnaCardio, AstraZeneca, Bayer, Berlin Cures, BMS, Cardurion, Cytokinetics, Ionis Pharmaceuticals, Novartis, Regeneron Pharmaceuticals, River 2 Renal Corp. Personal lecture fees: Abbott, Alkem Metabolics, Astra Zeneca, Blue Ocean Scientific Solutions Ltd, Boehringer Ingelheim, Canadian Medical and Surgical Knowledge, Emcure Pharmaceuticals Ltd, Eris Lifesciences, European Academy of CME, Hikma Pharmaceuticals, Imagica Health, Intas Pharmaceuticals, J.B. Chemicals & Pharmaceuticals Ltd, Lupin Pharmaceuticals, Medscape/HeartOrg, ProAdWise Communications, Radcliffe Cardiology, Sun Pharmaceuticals, The Corpus, Translation Research Group, Translational Medicine Academy. Data Safety Monitoring Board: WIRB-Copernicus Group Clinical Inc. He is a director of Global Clinical Trial Partners Ltd. The other authors report no conflicts.

Supplemental Material

Supplemental Methods
Tables S1–S15
Figures S1–S2

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